NETWORKING

ARP & ICMP Visual Guide

Finding neighbors, testing paths — the helper protocols of IP.

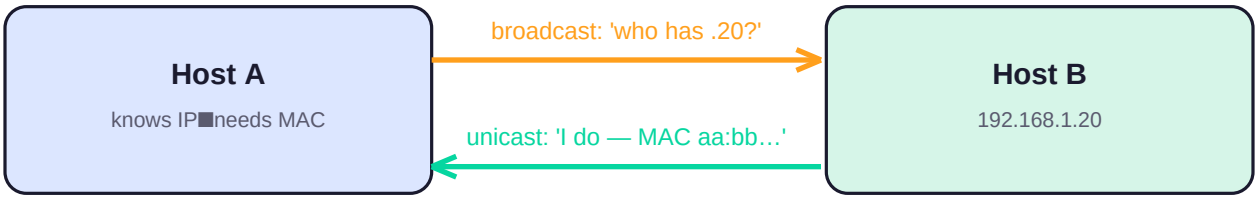
ANIMATED WALKTHROUGH

Step through pages to play the animation

STEP 1/5

ARP: Who Has This IP?

IP needs a MAC to send a frame — ARP asks the LAN.



- ARP request is broadcast — every host on the LAN hears it; only the owner replies.
- Results are cached in the ARP table (minutes) — repeat traffic needs no new broadcast.
- Proxy ARP: a router answers for distant hosts — useful, and a classic misconfiguration smell.

STEP 2/5

Gratuitous ARP

Announcing yourself without being asked.

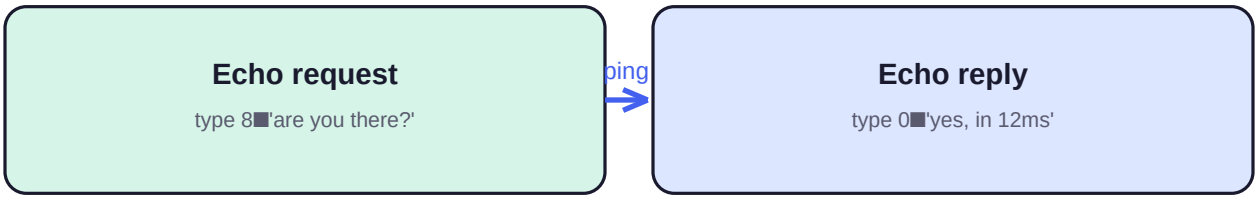
- A host broadcasts 'I am 192.168.1.20 at aa:bb:cc' unprompted — updates everyone's cache.
- Legit uses: failover (new active node claims the virtual IP), duplicate-address detection on boot.
- Dark side: ARP spoofing uses the same mechanism maliciously — switches fight back with Dynamic ARP Inspection.
- No authentication in ARP at all — on untrusted LANs, that's the vulnerability.

ARP is IPv4-only. IPv6 replaced it with Neighbor Discovery (NDP) over ICMPv6 — same job, better design.

STEP 3/5

ICMP: The Network's Messenger

Control messages about the network itself — not user data.



- Ping measures reachability + RTT — the first diagnostic for 'is it up?'.
- ICMP also carries: Destination Unreachable, Time Exceeded, Redirect, and PMTU 'packet too big'.
- Don't block all ICMP: killing 'packet too big' breaks Path MTU Discovery and causes mystery stalls.

STEP 4/5

Traceroute: TTL Tricks

Map the path by expiring packets on purpose.

TTL=1 → router 1 replies 'Time Exceeded' — hop 1 revealed

TTL=2 → dies at router 2 — hop 2 revealed

TTL=3 → dies at router 3 … until the destination answers

- Every router decrements TTL; at zero it sends ICMP Time Exceeded back — that reply IS the hop.
- Three probes per TTL: RTT variance per hop hints at congestion or asymmetric paths.
- Stars (* * *) = filtered ICMP or rate-limited routers — common, not necessarily failure.

STEP 5/5

Reading the Diagnostics

What ping and traceroute actually tell you.

- Ping loss with clean traceroute: endpoint or last-mile issue — not the path.
- Latency jump at one hop that persists downstream: that link is the bottleneck.
- Latency jump that VANISHES downstream: the router was slow to answer ICMP, not to forward — ignore it.
- Asymmetric paths mean forward and reverse can differ — diagnose both directions.

mtr blends both tools: continuous ping per hop — the fastest way to localize packet loss on a path.

STEP ✓I✓

Key Takeaways

ARP & ICMP — Remember

- ARP maps IP→MAC via broadcast request + cached replies; IPv6 uses NDP instead.
- Gratuitous ARP announces ownership — legit for failover, abused for spoofing.
- ICMP carries echo (ping), unreachable, time-exceeded, and MTU signals.
- Traceroute reveals hops by expiring TTLs on purpose; stars mean filtered ICMP.
- A latency spike that disappears downstream is slow ICMP, not a slow path.

Keep exploring

Pair this guide with the matching video walkthroughs and hands-on labs at interstitiumlabs.dev